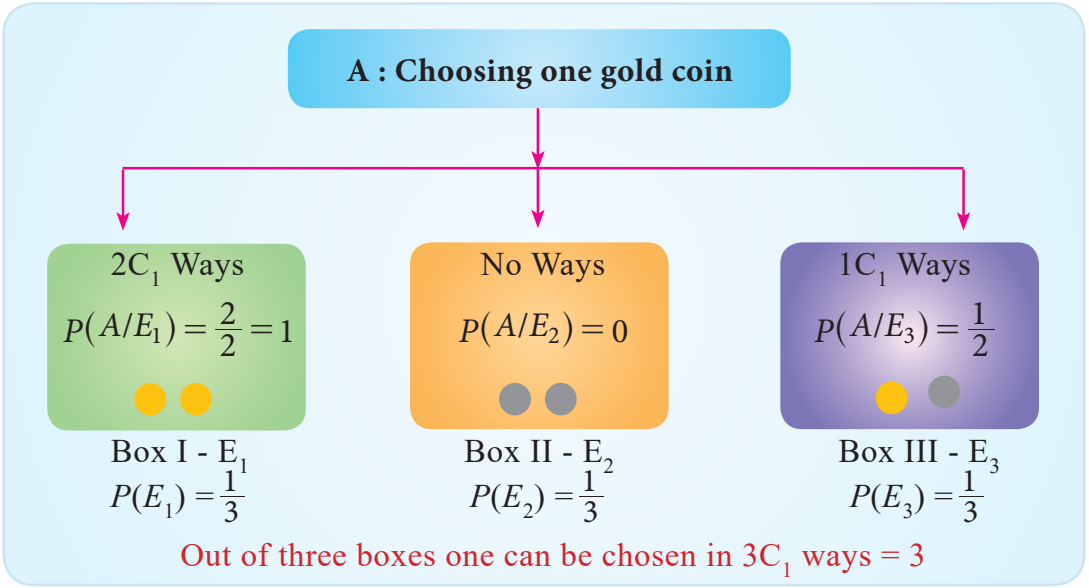


Fig. 8.11

If E_1, E_2 and E_3 be the events that the boxes I, II, III are chosen respectively.

Then $P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$

let A be the event that the gold coin is drawn

$P(A/E_1)$ = P (a gold coin from box 1) $= \frac{2}{2} = 1$

$P(A/E_2)$ = P (a gold coin from box 1I) $= \frac{0}{2} = 0$

$P(A/E_3)$ = P (a gold coin from box III) $= \frac{1}{2}$

Now, the probability that the other coin in the box is also gold is same as probability of choosing the box I and drawing a gold coin = $P(A/E_1)$

By Bayes' Theorem,

$$P(E_1/A) = \frac{P(E_1)P(A/E_1)}{\sum_{i=1}^{3} P(E_i)P(A/E_i)}$$

$$= \frac{\frac{1}{3} \times 1}{\frac{1}{3} \times 1 + \frac{1}{3} \times 0 + \frac{1}{3} \times \frac{1}{2}}$$

$$= \frac{2}{3}$$